PHYSICS

1) Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$. Which of the following statements is **INCORRECT** ?

(1) $\vec{A} \cdot \vec{B} = 50$

(2) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$

(3) $|\vec{A}| = 5$

(4) $\vec{A} \cdot \vec{B} = 48$

2) If $\vec{P} = 3\hat{i} + 4\hat{j}$ then find out the angle of $\vec{P}$ from x-axis.

(1) 60°

(2) 30°

(3) 37°

(4) 53°

3) If the three vectors $\vec{A}, \vec{B}, \vec{C}$ are coplaner then find n = ?

$\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$

$\vec{B} = n\hat{j} - 3\hat{k}$

$\vec{C} = 4\hat{i} + 2\hat{j} + 2\hat{k}$

(1) n = 2

(2) n = 12

(3) n = 3

(4) n = 1

4) What is the unit vector perpendicular to the following vectors $2\hat{i} + 2\hat{j} - \hat{k}$ and $6\hat{i} - 3\hat{j} + 2\hat{k}$:-

(1) $\frac{\hat{i} + 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$

(2) $\frac{\hat{i} - 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$

(3) $\frac{\hat{i} - 10\hat{j} - 18\hat{k}}{5\sqrt{17}}$

(4) $\frac{\hat{i} + 10\hat{j} + 18\hat{k}}{5\sqrt{17}}$

5) What is the angle between $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}$?

(1) $\frac{\pi}{6}$
(2) $\frac{\pi}{4}$
(3) $\frac{\pi}{3}$
(4) None of these

6) Vector which is perpendicular to $(a\cos\theta)\hat{i}+(b\sin\theta)\hat{j}$ is given by :-

(1) $(b\sin\theta)\hat{i}-(a\cos\theta)\hat{j}$
(2) $\frac{(\sin\theta)}{a}\hat{i}-\frac{(\cos\theta)}{b}\hat{j}$
(3) $5\hat{k}$
(4) All of these

7) Find the projection of $\vec{A}=\hat{i}-\hat{j}$ along $\vec{B}=-\hat{i}+\hat{j}$:-

(1) $\frac{-1}{\sqrt{2}}$
(2) $-\sqrt{2}$
(3) $\sqrt{2}$
(4) None of these

8) If $\vec{p}=3\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{Q}=\hat{i}+\hat{j}$ then find a vector whose direction is same as that of $\vec{Q}$ and magnitude is same as that of $\vec{p}$

(1) $\frac{5}{\sqrt{2}}(\hat{i}+\hat{j})$
(2) $5(\hat{i}+\hat{j})$
(3) $\frac{5}{\sqrt{2}}(3\hat{i}+4\hat{j}+5\hat{k})$
(4) $\frac{3\hat{i}+4\hat{j}+5\hat{k}}{5}$

9) The adjacent sides of a parallelogram are represented by the two vectors $\hat{i}+2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$. What is the area of parrelogram ?

(1) 8
(2) $8\sqrt{3}$
(3) $3\sqrt{8}$
(4) 192

10) A particle moves so that its position vector varies with time as $\vec{r}=A\cos\omega t\hat{i}+A\sin\omega t\hat{j}$. The

initial velocity of the particle is :

(1) $A\omega\hat{i}$
(2) $A\omega\hat{j}$
(3) $A\omega(\hat{i}+\hat{j})$
(4) $A\omega(\hat{i}-\hat{j})$

11) The position x of a particle varies with time, (t) as $x = at^2 - bt^3$. The acceleration will be zero at time t equal to :-

(1) $\frac{a}{3b}$
(2) Zero
(3) $\frac{2a}{3b}$
(4) $\frac{a}{b}$

12) A car travelling at a speed of 30 km/h is brought to rest in a distance of 8 m by applying brakes. If the same car is moving at a speed of 60 km/h then it can be brought to rest with same brakes in :-

(1) 64 m
(2) 32 m
(3) 16 m
(4) 4 m

13)

A boy moves along the curved path of quarter circle. Calculate the ratio of distance of displacement :-

(1) 11 : 7
(2) 7 : 11
(3) $\sqrt{2} \times 7 : 11$
(4) $11 : 7\sqrt{2}$

14)

A man travels half time of its journey with speed of 5 m/s and for remaining half time he moves half of its total distance travelled, with speed of 3 m/s & remaining half distance with speed 6 m/s. Then his average speed of man is :-

(1) 3 m/s
(2) 6 m/s
(3) 4 m/s
(4) 4.5 m/s

15)

A car is moving with speed 15 m/s. Suddenly the driver sees an obstruction on the road and takes 0.4 sec to apply the brake, which cause a deacceleration of 5 m/s^2. Then distance travelled by car before stopping :-

(1) 22.5 m

(2) 24.5 m

(3) 26.5 m

(4) 28.5 m

16)

A particle moves in the east direction with 15 m/sec for 2 sec then northwards with 5 m/s for 18 sec. Average speed of the particle is :–

(1) 1 m/s

(2) 5 m/s

(3) 6 m/s

(4) 10 m/s

17)

Starting from rest ,the acceleration of a particle is a = 2(t –1). The velocity of the particle at t = 6 sec. is :–

(1) 15 m/s

(2) 25 m/s

(3) 24 m/s

(4) none of these

18)

If a body starts from rest and travels 120m in the 8th second, then acceleration is :

(1) 16 m/s^2

(2) 10 m/s^2

(3) 0.227 m/s^2

(4) 0.03 m/s^2

19)

If a car covers $2/5^{th}$ of the total distance with v_1 speed and $3/5^{th}$ distance with v_2 then the average speed is

(1) $\frac{1}{2}\sqrt{v_1 v_2}$

(2) $\frac{v_1 + v_2}{2}$

(3) $\frac{2v_1v_2}{v_1+v_2}$

(4) $\frac{5v_1v_2}{3v_1+2v_2}$

20) If the distance covered is zero, the displacement :

(1) must be zero

(2) may or may not be zero

(3) cannot be zero

(4) depends upon the particle

21) A man walks 10 m towards east, then 40 m towards north and in the last $40\sqrt{2}$ m towards south-west. The displacement from origin is

(1) 30 m towards east

(2) $80\sqrt{2}$ m towards north-west

(3) 30 m towards west

(4) 20 m towards east-north

22) A car moves for half of its time at 80 km/hr and for rest half of time at 40 km/hr. Total distance covered is 60 km. What is the average speed of the car :

(1) 60 m/s

(2) 60 km/hr

(3) 120 km/hr

(4) 180 km/hr

23) A particle is moving along x-axis and it's position with time is given as $x(t) = 2t^2 - t - 1$, then choose incorrect statement (unit of t is second and that of x is metre) :-

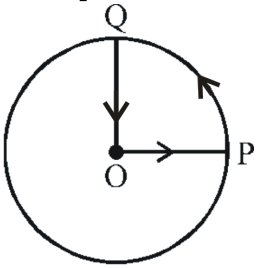
(1) Particle is at x = 0 at t = 1 sec

(2) Velocity of particle is zero at t = 0.25 sec

(3) While passing through origin, velocity of particle can be $3ms^{-1}$ (in + x direction)

(4) None

24) A cyclist starts from the centre O of a circular park of radius one kilometre, reaches the edge P of the park. Then cycles along the circumference and returns to the centre along QO as shown in the figure. If the round trip takes ten minutes, the net displacement and average speed of the cyclist (in metre and kilometre per hour respectively) is :-



(1) 0, 1

(2) $\frac{\pi + 4}{2}, 0$

(3) $21.4, \frac{\pi + 4}{2}$

(4) 0, 21.4

25) If the displacement of a particle varies with time as $\sqrt{x} = t + 3$:

(1) velocity of the particle is directly proportional to t^2

(2) velocity of particle varies linearly with t

(3) velocity of particle is proportional to $\sqrt{t}$

(4) initial velocity of the particle is zero

26)

A particle moves along a straight line such that its position at any time t is given by :

$s = (t^3 - 3t^2 + 2)m$ The position of the particle when the acceleration becomes zero is :-

(1) 0 m

(2) 2 m

(3) 3 m

(4) –2 m

27) A particle is moving under acceleration a = kt. The motion starts from rest. The velocity and displacement as a function of time t is :-

(1) kt^2, kt^3

(2) $\frac{kt^2}{2}, \frac{kt^3}{3}$

(3) $\frac{kt^3}{2}, \frac{kt^2}{3}$

(4) $\frac{kt^2}{2}, \frac{kt^3}{6}$

28) A particle moves, along a straight line OX. At a time t (in second) the distance x (in metre) of the particle from O is given by, $x = 20 + 12\ t - t^3$. How long would the particle travel before coming to rest ?

(1) 16 m

(2) 36 m

(3) 40 m

(4) 56 m

29) If a particle starts its motion in a straight line with an initial velocity of 10 m/s and retardation of 2 m/s^2 then find the distance covered by it in 4th second.

(1) 3 m

(2) 5/4 m

(3) 92 m

(4) 28 m

30) A particle moves in a straight line with a constant acceleration. It changes its velocity from $10\,ms^{-1}$ to $20\,ms^{-1}$ while passing through a distance of 135 m in t seconds. The value of t is:-

(1) 12

(2) 9

(3) 10

(4) 1.8

31) The velocity of a body depends on time according to the equation $v = 20 + 0.1t^2$. The body is undergoing:

(1) uniform acceleration

(2) uniform retardation

(3) non-uniform acceleration

(4) zero acceleration

32) A particle's position as a function of time is described as $y(t) = 2t^2 + 3t + 4$. What is the average velocity of the particle from t = 0 to t = 3 sec ?

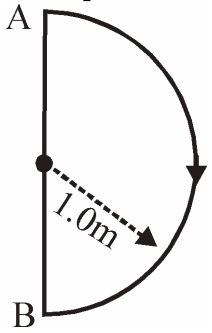
(1) 3 m/sec

(2) 6 m/sec

(3) 9 m/sec

(4) 12 m/sec

33) In 1.0 s, a particle goes from point A to B, moving in a semicircle of radius 1.0 m. The magnitude of the average velocity is :-



(1) 3.14 m/s

(2) 2.0 m/s

(3) 1.9 m/s

(4) Zero

34) For motion of an object along the x-axis, the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at x = 2m :-

(1) 48 m/s

(2) 80 m/s

(3) 18 m/s

(4) 10 m/s

35) A particle has initial velocity $(2\hat{i} + 3\hat{j})$ and acceleration $(0.3\hat{i} + 0.2\hat{j})$. The magnitude of velocity after 10 seconds will be :-

(1) 5 units

(2) 9 units

(3) $9\sqrt{2}$ units

(4) $5\sqrt{2}$ units

36) A particle is moving along a circular path of radius 6 m with a uniform speed of 8 ms^{-1}. The average acceleration when the particle completes one half of the revolution is:-

(1) $\frac{16}{3\pi} ms^{-2}$

(2) $\frac{32}{3\pi} ms^{-2}$

(3) $\frac{64}{3\pi} ms^{-2}$

(4) None of these

37) A body covers first 1/3 part of its journey with a velocity of 2 m/s, next 1/3 part with a velocity of 3 m/s and rest of the journey with a velocity 6m/s. The average velocity of the body will be:-

(1) 3 m/s

(2) $\frac{11}{3}$ m/s

(3) $\frac{8}{3}$ m/s

(4) $\frac{4}{3}$ m/s

38) A bullet looses $1/10^{th}$ of is velocity in passing through a plank. What is the least number of planks required to stop the bullet?

(1) 10

(2) 11

(3) 5

(4) 6

39) A drunkard is walking along a straight road. He takes 4 steps forward and 2 steps backward and so on. Each step is 1m long and takes 1s. There is a pit on the road 10 m away from the starting point. The drunkard will fall into the pit after.

(1) 29 sec

(2) 30 sec

(3) 24 sec

(4) 22 sec

40) A person first goes $10\sqrt{3}$ m in East direction. Then after he goes 10 m in north direction and finally he goes $20\sqrt{3}$ m 60° south of west. Net displacement of person will be :

(1) 20 m, in north

(2) 10 m, in north

(3) 20 m, in south

(4) 10 m, in south

41) An insect starts moving from origin at a constant velocity $\left(2\hat{i} + 3\hat{j}\right)$ m/s for 1 sec and then with constant velocity $\left(2\hat{i} + 2\hat{j}\right)$ m/s for $\frac{1}{2}$ sec. The distance of the insect from origin after these displacements is

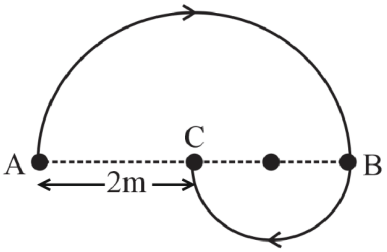
(1) 4 m

(2) 5 m

(3) 7 m

(4) 10 m

42) Find ratio of distance to displacement as the particle goes from A to C along the curve shown



(AC = CB)

(1) $\frac{\pi}{2}$

(2) π

(3) $\frac{3\pi}{2}$

(4) 2π

43) A car accelerate from rest at a constant rate of 2 m/s^2 for some time. Then, it retards at constant rate of 4 m/s^2 & comes to rest. If it remain in motion for 3 sec, then the maximum speed attained by the car is :-

(1) 2 m/s

(2) 3 m/s

(3) 4 m/s

(4) 6 m/s

44) If the velocity of a particle is given by $v = (180 - 16x)^{1/2}$ m/s, then its acceleration will be:-

(1) Zero

(2) 8 m/s^2

(3) –8 m/s^2

(4) 4 m/s^2

45) A bullet fired into a fixed target loses half of its velocity after penetrating 3 cm. How much further it will penetrate before coming to rest assuming that it faces constant resistance to motion ?

(1) 1.5 cm

(2) 1.0 cm

(3) 3.0 cm

(4) 2.0 cm

CHEMISTRY

1) 10^{-2} mm = ...?.... m

(1) 10

(2) 10^{-5}

(3) 10^{5}

(4) 1

2) Find the **incorrect** relation?

(1) 1 amu = 1.67×10^{-24} g

(2) 1 eV = 1.6×10^{-19} J

(3) 1 erg = 10^{7} J

(4) 1 atm = 760 torr

3) 4 cm^3 = _____ mL

(1) 4000

(2) 4×10^{-3}

(3) 4

(4) 400

4) The number of molecules in 16 g of methane (CH_4) is:-

(1) 3.0×10^{23}

(2) 6.02×10^{23}

(3) $\frac{16}{6.02} \times 10^{23}$

(4) $\frac{16}{3.0}\times10^{23}$

5) 7.5 grams of a gas occupy 6 litres of volume at STP the gas is

(1) NO
(2) N_2O
(3) CO
(4) CO_2

6) What is the mass of 3.01×10^{22} molecules of ammonia?

(1) 8.5 g
(2) 0.85 g
(3) 0.425 g
(4) 4.25 g

7) A gas is found to have the formula $(CO_2)_x$. It's vapour density is 44. The value of x must be :-

(1) 2
(2) 1
(3) 4
(4) 3

8) Which of the following pair of gases contains the same number of molecules?

(1) 16g of O_2 and 14g of N_2
(2) 8g of O_2 and 22g of CO_2
(3) 28g of N_2 and 22g of CO_2
(4) 32g of O_2 and 32g of N_2

9) Total no. of electrons present in 48g Mg^{2+} are:

(1) 24 N_A
(2) 2 N_A
(3) 20 N_A
(4) 10 N_A

10) Which of the following has minimum number of atoms of oxygen:-

(1) 10 mL $H_2O_{(\square)}$
(2) 0.1 mol $V_2O_{5(s)}$
(3) 12 g of $O_{3(g)}$
(4) 12.04×10^{22} molecules of CO_2

11) The mass of 1 molecule of water is approximately :-

(1) 18 g
(2) 3×10^{-23} g
(3) 1.5×10^{-23} g
(4) 4.5×10^{-23} g

12) If 6×10^{22} molecules are removed from 120 g of $CaCO_3$. then find out number of moles of $CaCO_3$ are left.

(1) 1.3 mol
(2) 1.2 mol
(3) 1 mol
(4) 1.1 mol

13) A gaseous mixture contains O_2 and N_2 in the ratio of 1 : 4 by weight. The ratio of their number of molecules is :-

(1) 1 : 4
(2) 1 : 8
(3) 7 : 32
(4) 3 : 16

14) Actual mass of CO_2 is :-

(1) 44 g
(2) 44 kg
(3) $44 \times 1.67 \times 10^{-24}$ g
(4) 1.67×10^{-24} g

15) Which contains maximum no of atoms :-

(1) 16 g 'O'
(2) 16 g 'O_2'
(3) 16 g 'O_3'
(4) All will have same number of atoms

16) Aspirin has the formula $C_9H_8O_4$. How many atoms of oxygen are there in a tablet weighing 360 mg?

(1) 1.204×10^{23}
(2) 1.08×10^{22}
(3) 1.204×10^{24}
(4) 4.81×10^{21}

17) An organic compound contains 4% sulphur. Its minimum molecular weight is :

(1) 200
(2) 400
(3) 800
(4) 1600

18) How many mL of 0.1 M NaOH contains 0.4 g NaOH ?

(1) 1000 mL
(2) 10 mL
(3) 100 mL
(4) 10^4 mL

19)

1.5 moles of O_2 combine with Mg to form MgO. The moles of Mg that has combined is

$2Mg + O_2 \rightarrow 2MgO$
(1) 3 mol
(2) 1.5 mol
(3) 2 mol
(4) 1 mol

20) What volume of O_2 gas will required for the complete combustion of 10 L of CH_4 gas :-

(1) 10 L
(2) 30 L
(3) 20 L
(4) 15 L

21) What amount of CaO will be produced by 2 moles of Ca.
$2Ca + O_2 \rightarrow 2CaO$

(1) 5.6g
(2) 56g
(3) 112g
(4) 224g

22) 6.02×10^{20} molecules of Urea are present in 100 mL solution. Find Molarity of solution.

(1) 10^{-1} M
(2) 10^{-2} M
(3) 10^{-3} M
(4) None

23) What volume of CO_2 will be formed by complete combustion of 2L of C_3H_8.

(1) 3 L

(2) 6 L

(3) 5 L

(4) 10 L

24) Volume of air needed for complete combustion of 10 litres of C_2H_4 is :
$C_2H_{4(g)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 2\ H_2O(g)$

(1) 30 litres

(2) 50 litres

(3) 90 litres

(4) 150 litres

25) In the reaction : $2A\square + 3Cl_2 \rightarrow 2AlCl_3$
If 54 g of $A\square$ reacts with 71 g of Cl_2, find the limiting reagent.

(1) Al

(2) Cl_2

(3) Both react completely

(4) None

26) A compound contains 40% sulfur and 60% oxygen by mass. Find its empirical formula.
(Atomic masses : S = 32, O = 16)

(1) SO_2

(2) SO_3

(3) S_2O_3

(4) S_2O_2

27)

A hydrocarbon contains 85.7% carbon and 14.3% hydrogen. Vapour density = 28. Find molecular formula. (C = 12, H = 1)

(1) C_2H_4

(2) C_4H_8

(3) C_3H_6

(4) C_2H_6

28) For a reaction $2A + 3B \rightarrow 2C$, the amount of C formed by starting the reaction with 4 moles of A and 12 moles of B is :-

(1) 8 mol

(2) 4 mol

(3) 2 mol

(4) 1 mol

29) How many grams of CO_2 will be produced on the complete decomposition of 40g of 10% pure $CaCO_3$?

(1) 176 g
(2) 17.6 g
(3) 1.76 g
(4) 0.176 g

30) 10 mol SO_2 and 15 mol O_2 were allowed to react the remain moles of SO_2 and O_2 respectively are:
$2SO_2 + O_2 \rightarrow 2SO_3$

(1) 0 mol, 10 mol
(2) 2 mol, 8 mol
(3) 4 mol, 5 mol
(4) 8 mol, 2 mol

31) If 2 moles of A is present in 3 moles of B. Then calculate the mole fraction of A :-

(1) 0.4
(2) 0.6
(3) 0.3
(4) 0.7

32) How much mass of Fe will be required in below process to produce 11.2 L of H_2 at STP?
(At. wt. of Fe = 56 amu)
$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

(1) 5.6 g
(2) 2.8 g
(3) 1.88 g
(4) 21g

33) How many moles of lead nitrate is needed to produce 224 litre of oxygen at N.T.P.?
$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

(1) 2 mol
(2) 20 mol
(3) 10 mol
(4) 5 mol

34) Total number of neutrons present in 54 mL of water is at $25^{\circ}C$ is

(1) $3\ N_A$
(2) $30\ N_A$
(3) $24\ N_A$
(4) $15\ N_A$

35) 1 g of hydrogen reacts with 4 g of oxygen to form water. The mass of water formed is

(1) 18
(2) 5 g
(3) 4.5 g
(4) 2.5 g

36) An metal oxide of metal 'M' has 40% by mass of oxygen. Metal 'M' has atomic mass is 24. The empirical formula of the oxide is

(1) M_2O
(2) M_2O_3
(3) MO
(4) M_3O_4

37) In this reaction,
$2A\square + \frac{3}{2} O_2 \rightarrow A\square_2O_3$,
9 gram of $A\square$ will react with :

(1) 6 gram O_2
(2) 9 gram O_2
(3) 4 gram O_2
(4) 8 gram O_2

38) 18 gm of glucose present in 162 gm of H_2O. Calculate mass fraction of glucose:-

(1) 0.02
(2) 0.1
(3) 0.3
(4) 0.5

39) **Assertion :** The amount of the products formed in a reaction depends upon the limiting reagent.
Reason : Limiting reactant reacts completely in the reaction therefore limits amount of product.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

40)

Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R) .
Assertion (A) : Equal volume of all gases contains equal no. of atoms, under same conditions of P and T.
Reason (R) : Atom is the smallest particle which takes part in chemical reactions.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is incorrect, but (R) is correct.

(4) Both (A) & (R) are incorrect.

41) For endothermic reaction :-

(1) ΔH = positive

(2) ΔH = negative

(3) Both (1) and (2)

(4) None of these

42) A reaction is said to be in equilibrium if :-

(1) Concentration of reactant and product becomes equal.

(2) 50% reactant convert into product

(3) Rate of forward and backward reaction becomes equal.

(4) All of the above

43) **Assertion:-** Addition of catalyst to a reaction mixture, increases the amount of reactants and products at equilibrium.
Reason:- The forward reaction becomes slower on adding catalyst.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True and the Reason is False.

(4) Both Assertion & Reason are False.

44) Which of the following is not true about equilibrium :

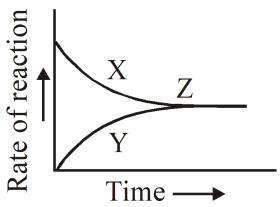
(1) Rate of forward reaction equals to rate of backward reaction

(2) It is static in nature

(3) It is not affected by the presence of catalyst

(4) At equilibrium, concentration is constant

45) Study the given figure and label X, Y and Z.



	X	Y	Z

(1)	Backward reaction	Forward reaction	Products
(2)	Forward reaction	Backward reaction	Equilibrium
(3)	Reversible reaction	Irreversible reaction	Equilibrium
(4)	Forward reaction	Forward reaction	Backward reaction

(1) 1

(2) 2

(3) 3

(4) 4

BIOLOGY

1) Anyone of the following set of characters is definitely found at some stage in the life cycle of a chordate?

(1) Mammary gland, hairs, gill cleft

(2) Notochord, vertebral coloum, gill cleft

(3) Notochord, pharyngeal gill cleft, dorsal tubular nervous system

(4) Dorsal tubuler nervous system, vertebral coloumn, scales

2) Common characteristics found in between aves and mammals are :-
(A) Respiration occurs by lungs
(B) Heart is four chambered
(C) Homeothermy
(D) Larva is not found
(E) Dioecious and internal fertilisation
(F) Jaws are provided with teeth

(1) 3

(2) 4

(3) 5

(4) 6

3) Which of the following is not a fundamental characteristic of chordata phylum ?

(1) Notochord

(2) Dorsal hollow nerve cord

(3) Cranium

(4) Post anal tail

4) **Statement I :** In Cyclostomata, cranium and vertebral column are composed of cartilage.
Statement II : *Exocoetus* and *Clarias* belong to class cyclostomata

(1) Statements I and II are correct

(2) Statements I and II are incorrect.

(3) Statement I is correct but II is incorrect

(4) Statement II is correct and I is incorrect

5) Which of the following characters is not present in all chordates?

(1) Diaphragm

(2) Coelom

(3) Pharyngeal gill clefts

(4) Dorsal nerve cord

6) Find out incorrect match from column I, II and III :-

	Column-I	**Column-II**	**Column-III**
(1)		*Petromyzon*	Scales absent
(2)		*Pristis*	Air bladder absent but operculum is present
(3)		*Hippocampus*	Air bladder and operculum both are present
(4)		*Rana*	Skin is scaleless and tympanum represents the ear

(1) 1

(2) 2

(3) 3

(4) 4

7) Heart is three chambered in reptiles except

(1) Turtle

(2) Chameleon

(3) Cobra

(4) Crocodile

8) Which one of the following is incorrect for Aves?

(1) Heart is four-chambered and animals are oviparous

(2) The presence of air cavities in bones and the presence of feathers on the body

(3) Digestive tract has additional chambers crop and gizzard and animals are homeothermous

(4) The forelimbs have scales

9) How many statements are correct ?
(i) Cephalochordates are exclusively marine.
(ii) Notochord in cephalochordates in present only during larval stages.
(iii) Salpa is a member of cephalochordata

(1) 1

(2) 2

(3) 3

(4) 2 & 3

10) Which of the following is correct about cyclostomes?

(1) Ectoparasite on some fishes.

(2) Skin with scales and contain unicellular mucous gland.

(3) Larva is marine

(4) Heart is two chambered and possess 4 pair of gill slits for respiration.

11) First class of vertebrate, which are fully adapted for terrestrial life is -

(1) Amphibia

(2) Reptilia

(3) Aves

(4) Mammals

12) Which of the following is incorrect about reptilia?

(1) Body is covered by dry and cornified skin, epidermal scales or scutes.

(2) Body is divisible into head, neck, trunk and tail.

(3) Heart is usually three chambered.

(4) Respiration occur through lungs and gills in aquatic snakes.

13) Select the incorrect matching:

(1)	Chelone	Turtle
(2)	Testudo	Tortoise
(3)	Chameleon	Tree lizard
(4)	Calotes	Wall lizard

(1) 1

(2) 2

(3) 3
(4) 4

14) Which one of these animals is not a homeotherm?

(1) Camelus
(2) Chelone
(3) Macropus
(4) Pisittacula

15)

Match the following columns.

	Column-I		**Column-II**
(A)	Cyclostomes	(1)	Hemichordata
(B)	Aves	(2)	Urochordata
(C)	Tunicates	(3)	Agnatha
(D)	Balanoglossus	(4)	Pisces
(E)	Osteichthyes	(5)	Tetrapoda

Codes:

	A	**B**	**C**	**D**	**E**
(1)	1	2	3	4	5
(2)	2	3	4	1	5
(3)	3	5	2	1	4
(4)	3	1	5	2	4

(1) 1
(2) 2
(3) 3
(4) 4

16) Which one is not cartilaginous fish?

(1) Carcharoden (great white shark), Trygon (sting ray)
(2) Exocoetus (flying fish), catla (katla), clarius (Mangur)
(3) Scolidon (dog fish)
(4) Pristis (saw fish)

17) Which of the following is not a characteristic of class chondrichthyes?

(1) Gill slits are separated and without operculum.
(2) Predaceous with powerful jaws.

(3) Notochord is persistent throughout life.

(4) Airbladder present.



18) Identify the incorrect statement regarding the class of given organism

(1) Tympanum represents ear

(2) Fertilisation is internal

(3) Three chambered heart is found in all members

(4) Despite being tetrapods, limbs may not be present in every member of this class

19) Find out the wrong difference between chordates and nonchordates from the table given below :-

	CHORDATES	**NONCHORDATES**
(1)	Notchord present	Notochord absent
(2)	Heart is dorsal	Heart is ventral
(3)	Dorsal, hollow nerve cord is present	Ventral, solid nerve cord is present
(4)	Post anal tail is present	Post anal tail is absent

(1) 1

(2) 2

(3) 3

(4) 4

20) ___ is/are marine but migrate for spawning to fresh water :-

(1) Lamprey

(2) Myxine and Torpedo

(3) Lamprey and Torpedo

(4) Torpedo and Trygon

21) Read the following characters carefully :-
(a) Placoid scales are present
(b) Mouth is located ventrally
(c) Air bladder is absent

These characters are found in

(1) Scoliodon
(2) Catla
(3) Exocoetus
(4) Labeo

22) How many of the following show vivipary?
Pteropus, Camelus, Delphinus,Macropus, Ornithorhynchus, Neophron, Aptenodytes, Rattus, Equus

(1) 6
(2) 5
(3) 8
(4) 9

23) Which of the following is not the characteristic feature of amphibians?

(1) body is divisible into head and trunk
(2) moist skin
(3) oviparity and direct development
(4) eyes have eyelids and tympanum present

24) Which of the following is not the feature of reptiles?

(1) external ear absent / tympanum present
(2) heart usually three chambered / four chambered in crocodile
(3) oviparity / indirect development
(4) poikilothermous

25) Presence of feathers, beak, pneumatic bones, crop, gizzard, homeothermal, Oviparity, external fertilization
How many characters are related to Aves?

(1) 4
(2) 5
(3) 7
(4) 6

26)

(a) Streamlined body
(b) Four pair of gills
(c) Air bladder is present
(d) In males pelvic fins bear claspers
(e) Two chambered heart
(f) Fertilization is usually external
How many statements are correct for the members of class osteichthyes?

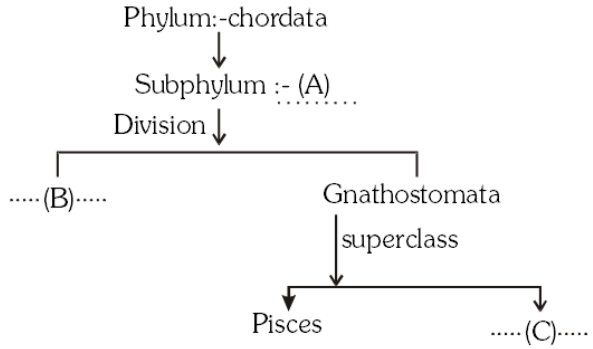
(1) one

(2) two

(3) five

(4) six

27) Given below is the chart of classification of chordata in which some group are left as blank and represented through alphabates A, B and C. Identify A, B and C and select the correct option?



(1) A = Protochordata, B = Agnatha, C = Tetrapoda

(2) A = Vertebrata, B = Agnatha, C = Tetrapoda

(3) A=Vertebrata, B= Cephalochordata, C = Tetrapoda

(4) A = Craniata, B = Vertebrata, C = Agnatha

28) **Assertion :** Acraniata is a group of organisms which do not have distinct cranium.
Reason : It includes small marine forms without head.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) The assertion is true but the reason is false.

(4) Both the assertion and reason are false.

29) You are observing an animal closely and you find some characters such as :-
* Endoskeleton made up of cartilage.
* Mouth is located ventrally.
* The skin is tough, containing placoid scales .
* Notochord persist through out life.
* Air bladder is absent.
After observing the above features, in which group will you classify that animal ?

(1) Chondrichthyes

(2) Osteichthyes

(3) Cyclostomata

(4) Cephalochordata

30) Which one of the following groups of animals is **correctly** matched with its one characteristic feature without even a single exception ?

(1) Mammalia - give birth to young ones

(2) Reptilia - possess 3-chambered heart with one incompletely divided ventricle

(3) Chordata - possess a mouth provided with an upper and lower jaw

(4) Chondrichthyes - possess cartilaginous endoskeleton

31) Mark the following statements as true/false.
A. The pharynx serves as a common passage for both food and air.
B. The larynx is a cartilaginous box that aids in sound production.
C. The trachea extend upto mid thoracic cavity.
D. The sound box refers to the nasal chamber.

(1) T, T, T, F

(2) F, F, T, F

(3) F, F, F, T

(4) T, T, F, F

32) Structures like the trachea, primary, secondary, and tertiary bronchi, and initial bronchioles are structurally supported by _____.
Fill in the blanks with appropriate words :

(1) complete muscular rings

(2) incomplete cartilaginous rings

(3) bony plates

(4) fibrous tissue bands

33) The fluid found between the double layers covering the lungs serves to _____ during breathing movements.
Fill in the blanks with appropriate words.

(1) increase surface tension

(2) provide nutrients

(3) reduce friction

(4) facilitate gas exchange

34) Functional residual capacity in human is the amount of air :

(1) that can be filled in lungs by forceful inspiration.

(2) that can be breathed out after forceful expiration.

(3) that remains in the lungs after normal expiration.

(4) that remains in the lungs after forceful expiration.

35) Most oxygen is carried by the blood __(i)__. Most carbon dioxide is carried by the blood __(ii)__.
Mark the correct option w.r.t (i) and (ii)

(1) (i) – as oxyhaemoglobin,
(ii) – in the form of bicarbonate ions

(2) (i) – dissolved in plasma,
(ii) – dissolved in plasma

(3) (i) – in the form of H^+-ions,
(ii) – in the form of bicarbonate ions

(4) (i) – attached to haemoglobin,
(ii) – attached to haemoglobin

36) Which of the following are the stages of respiration in correct order?
(A) Gaseous transport
(B) Cellular respiration
(C) Gaseous exchange at alveolar level
(D) Breathing

(1) A – D – C – B

(2) D – A – C – B

(3) D – A – B – C

(4) D – C – A – B

37) Match the column I with column II :

	Column-I		**Column-II**
A	Tracheal tubes	I	Scorpion
B	Gills	II	Frog
C	Moist skin	III	Insect
D	Book lungs	IV	Aquatic arthropods

	A	**B**	**C**	**D**
(1)	III	IV	II	I
(2)	III	I	IV	II
(3)	IV	III	II	I
(4)	IV	II	I	III

(1) 1

(2) 2

(3) 3

(4) 4

38) Respiration in insects is called direct because

(1) Tracheal tubes exchanges O_2 and CO_2 directely between air and tissue/cells.

(2) The tissues exchange O_2 / CO_2 directly with coelomic fluid

(3) The tissue exchange O_2 / CO_2 directly with the air outside through body surface

(4) Tracheal tubes exchange O_2 / CO_2 directly with the haemocoel which then exchange with tissues

39) What will be the P_{O_2} and P_{CO_2} in the atmospheric air compared to those in the alveolar air ?

(1) P_{O_2} lesser, P_{CO_2} higher

(2) P_{O_2} higher, P_{CO_2} lesser

(3) P_{O_2} higher, P_{CO_2} higher

(4) P_{O_2} lesser, P_{CO_2} lesser

40) ***Assertion :-*** Larynx is a cartilaginous box which helps in sound production.
Reason :- Epiglottis flap covers the glottis during swallowing and prevents the entry of food into the larynx.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

41) ***Assertion (A) :-***The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung cavity. (Pulmonary cavity) ***Reason (R) :-*** Lungs have no muscles so we cannot directly alter the pulmonary volume.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct but (R) is the correct explanation of (A)

42) **Assertion (A) :** Role of oxygen in regulation of breathing is quite insignificant.
Reason (R) : Chemosensitive receptors are more sensitive to pCO_2 and H^+ ions than pO_2.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

43) Read the following statements–
A. High H^+ concentration and higher temperature are favourable for dissociation of oxygen from oxyhaemoglobin.
B. Partial pressure of oxygen in alveoli is higher than in tissue.
C. The maximum volume of air a person can breathe in, after a forced expiration is vital capacity.
D. In the alveoli, where there is high pO_2, low pCO_2, lesser H^+ concentration and lower temperature.
How many factors are all favourable for the formation of oxyhaemoglobin ?

(1) One

(2) Two

(3) Three

(4) Four

44)

Arrange the following value in decreasing order:-
(a) Inspiratory reserve volume
(b) Inspiratory capacity

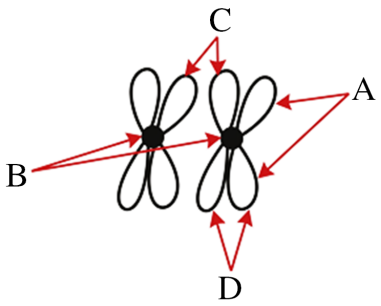
(c) Vital capacity
(d) Total lung capacity

(1) (d), (b), (c), (a)
(2) (a), (b), (c), (d)
(3) (d), (c), (b), (a)
(4) (d), (b), (a), (c)

45) What happens to O_2-Hb dissociation curve if CO_2 level of blood rises at 37°C ?

(1) No change occurs
(2) Becomes hyperbolic
(3) Shifts towards right
(4) Shift towards left

46) ln the given figure, which of the following represent nonsister chromatids?



(1) A
(2) B
(3) C
(4) D

47) Identify the incorrectly described chromosome type:

(1) Metacentric – equal arms
(2) Sub-metacentric – unequal arms
(3) Acrocentric – one very short arm
(4) Telocentric – centromere absent

48) The 80S eukaryotic ribosome consists of:

(1) 50S and 30S
(2) 60S and 40S
(3) 40S and 40S
(4) 50S and 40S

49) What is the membrane-covered core of a cilium or flagellum called?

(1) Radial spoke

(2) Axoneme
(3) Central sheath
(4) Centriole

50) The disc-shaped structures present on the sides of the centromere, which serve as attachment points for spindle fibers, are called _____.

(1) Chromatids
(2) Kinetochores
(3) Satellites
(4) Telomeres

51) Which protein is the major structural component of centrioles?

(1) Actin
(2) Myosin
(3) Tubulin
(4) Dynein

52) The primary function of nuclear pore complex is:

(1) DNA replication
(2) Transcription
(3) Selective nucleocytoplasmic transport
(4) Lipid synthesis

53) 9+2 arrangement of microtubules is seen in:

(1) Centriole
(2) Basal body
(3) Cilia and flagella
(4) Spindle fibres

54) Which organelle is commonly referred to as the ‘protein factory’ of the cell ?

(1) Golgi apparatus
(2) Nucleus
(3) Ribosome
(4) Endoplasmic reticulum

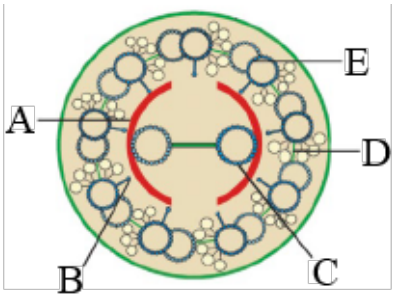
55) The shorter and longer arms of a submetacentric chromosome are referred to as:

(1) s-arm and l-arm respectively
(2) p-arm and q-arm respectively
(3) q-arm and p-arm respectively
(4) m-arm and n-arm respectively

56) The chromosomes in which centromere is situated close to one end are:

(1) Acrocentric
(2) Telocentric
(3) Sub-metacentric
(4) Metacentric

57) Identify the A, B, C, D and E ?



	A	B	C	D	E
(1)	Central microtubules	Radial spoke	Central sheath	Inter doublet bridge	Peripheral microtubules
(2)	Central sheath	Interdoublet bridge	Central sheath	Radial spoke	Peripheral microtubules
(3)	Central sheath	Radial spoke	Central microtubule	Inter doublet bridge	Peripheral microtubules
(4)	Central microtubules	Radial spoke	Central sheath	Peripheral bridge	Inter doublet bridge

(1) 1
(2) 2
(3) 3
(4) 4

58) Chromatin contains:-

(1) DNA + RNA only
(2) DNA + Acidic histone proteins + Non - histone + RNA
(3) DNA + basic histone protein + Non - histone protein only
(4) DNA + basic histone protein + Non - histone protein + RNA

59) ds circular DNA in plant cells are observed in :

(1) Mitochondria
(2) Chloroplast
(3) Nucleus
(4) Both (1) & (2)

60) Which one of the following statement is/are not correct for 80s ribosome?
(a) Present in prokaryote and eukaryote
(b) Not surrounded by any membrane
(c) "S" is indirectly is a measurement of size and density
(d) Composed of two subunit 50s and 30s

(1) Only a
(2) Only b & c
(3) Only a and d
(4) Only a & c

61) The telomeres of eukaryotic chromosomes consist of short sequences of:

(1) Adenine rich repeats
(2) Guanine rich repeats
(3) Thymine rich repeats
(4) Cytosine rich repeats

62) Read following statements:
(A) Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.
(B) The content of nucleolus is continous with the rest of the nucleoplasm as it is a membrane bound structure
(C) Nucleolus is main site for synthesis of ribosomal protein.
(D) Nuclear envelope consists of two parallel membranes with a space between called perinuclear space.
(E) The centriole form the basal body of cilia or flagella.
Choose the option which have all the **correct** statements :-

(1) A, B, D
(2) B, D, E
(3) A, D, E
(4) A, C, E

63)

Which of the following are functions of cytoskeleton?

(1) Motility
(2) Mechanical support
(3) Maintenance of the shape of the cell
(4) All of the above

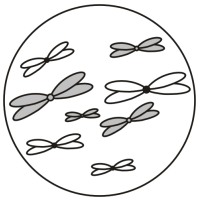
64) The number of mitochondria per cell is depending on the :-

(1) anatomy of the cells
(2) morphology of the cells
(3) physiological activity of the cells

(4) number of chromosomes in the cells

65) Read the following statements :-
(A) Both the chromatids of chromosome are joined by a structure called centromere
(B) During cell division spindle fibres are not formed
(C) Telomeric chromosome has a terminal centromere
(D) Every chromosome essentially has a secondary constriction
Find out correct sequence of true (T) and false (F) statements respectively :

(1) T F T T
(2) F T F T
(3) T F T F
(4) T T F F



66) Which of the following option correctly described the cell as shown in figure ?

(1) 8 chromosomes ; 20 homologous pair
(2) 16 chromosomes ; 4 homologous pair
(3) 8 chromosomes ; 4 homologous pair
(4) 16 chromosomes ; 8 homologous pair

67) Which of the following is true for nucleolus ?

(1) It is a double membrane bound structure.
(2) It takes part in spindle formation.
(3) It is a site for active ribosomal RNA synthesis
(4) It involve in oxidation of fats.

68) Match List - I with List - II.

	List - I		**List - II**
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below :

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(iv)	(i)	(ii)
(2)	(iii)	(ii)	(iv)	(i)

(3)	(iv)	(iii)	(ii)	(i)
(4)	(i)	(iv)	(iii)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

69) Match plastids with stored nutrients:

Column I (Plastid)	**Column II (Stored Nutrient)**
(a) Amyloplasts	(i) Proteins
(b) Elaioplasts	(ii) Carbohydrates (starch)
(c) Aleuroplasts	(iii) Oils and fats

(1) a-ii, b-i, c-iii

(2) a-ii, b-iii, c-i

(3) a-iii, b-ii, c-i

(4) a-i, b-iii, c-ii

70) **Assertion (A):** Mitochondria are called power houses of the cell.
Reason (R): They synthesise ATP during aerobic respiration.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

71) **Statement I:** Cristae increase surface area of inner mitochondrial membrane.
Statement II: Cristae are formed by infoldings of outer membrane.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

72) **Statement I:** Cytoskeleton provides mechanical support to the cell.
Statement II: Cytoskeleton is composed only of microtubules.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

73) Spherosomes are involve in–

(1) Synthesis of protein
(2) Synthesis and storage of lipid
(3) β-oxidation of fatty acid
(4) Storage of charbohydrate

74) Steroidal hormones are synthesised by :-

(1) RER
(2) Lysosome
(3) SER
(4) Ribosome

75) Peroxisomes have-

(1) ribosome
(2) DNA
(3) catalase enzyme
(4) centrosome

76) In which phase cells remain metabolically active but no longer divide :-

(1) S
(2) G_2
(3) Prophase
(4) G_0

77) In higher plant cell, during the S-phase DNA replication begins in the nucleus and :-

(1) the centriole duplicates in the cytoplasm
(2) histone protein synthesised
(3) the number of DNA molecules remains same
(4) all proteins of the cell are synthesised

78) The most dramatic period of the cell cycle :-

(1) Interphase
(2) S-phase
(3) M-phase
(4) G_1-phase

79) In which phase of cell cycle, cell is metabolically active and continuously grows but DNA content does not double ?

(1) G_1
(2) G_2

(3) S-phase

(4) Metaphase

80) The centriole undergoes dupication during....(i)...of cell cycle and begins to move towards opposite pole of the cell during the...(ii)... of cell cycle. Choose the correct option :-

(1) (i) Interphase (ii) M phase

(2) (i) G_2 phase (ii) Anaphase

(3) (i) G_1 phase (ii) Metaphase

(4) (i) Anaphase (ii) Prophase

81) Which of the following statement is incorrect:

(1) Cell growth is a continuous process while DNA synthesis occurs only during one specific stage in the cell cycle

(2) Duration of cell cycle can vary from organism to organism and also from cell type to cell type

(3) Cell at the G_0 phase is metabolically inactive but no longer proliferate unless called on to do so depending on the requirement of the organism

(4) M-phase is the most dramatic period of the cell cycle.

82) First gap phase in cell cycle is :-

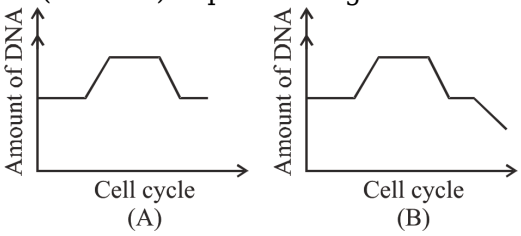
(1) interval between mitotic phase and initiation of DNA replication

(2) interval between end of DNA replication and DNA separation phase

(3) interval between karyokinesis and cytokinesis

(4) interval between DNA replication phase and second gap phase

83) The following graphs (A and B) depicts changes in DNA Content during cell cycle. Identify the cell division A and B :-



(1) A → Meiosis–II, B → Meiosis–I

(2) A → Mitosis, B → Meiosis–I

(3) A → Meiosis–I, B → Mitosis

(4) A → Mitosis, B → Meiosis

84) Interphase subdivided in to

(1) M, G, S, G_2

(2) G_1, S, G_2

(3) G_1, G_0, G_2, M

(4) S, G_2, M

85) Spindle fibres are composed of :

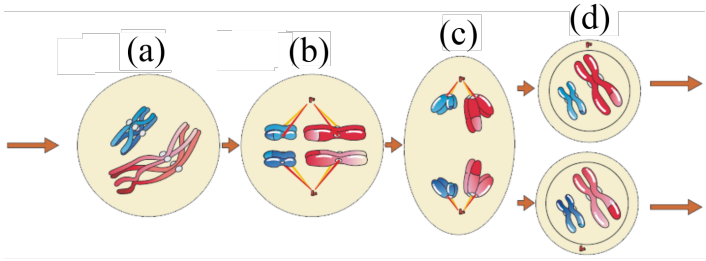
(1) Collagen
(2) Actin
(3) Spectrin
(4) Tubulin

86) A cell has 2n = 42 chromosomes then what will be number of chromosome during metaphase :-

(1) 21
(2) 42
(3) 94
(4) 44

87) Which stage of mitosis is marked by alignment of chromosomes at the equatorial plate?

(1) Prophase
(2) Anaphase-I
(3) Telophase-II
(4) Metaphase

88) 

On the basis of above diagrams the study of morphology of chromosomes can be easily studied in

(1) a
(2) b
(3) c
(4) d

89) Which of the following event is not observed during meiosis–I ?

(1) Pairing of homologous chromosomes
(2) Segregation of homologous chromosomes
(3) Crossing over
(4) Division in centromere

90) Which pair is not true?

(1) Synapsis → Zygotene
(2) Crossing over → Pachytene (Gene-exchange)
(3) Diplotene → Longest in oogenesis

(4) Meiosis division → 2 times S-phase

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	3	3	4	4	2	2	2	2	1	2	4	4	4	3	3	1	4	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	4	4	2	1	4	1	1	2	3	3	2	2	4	3	1	4	4	3
Q.	41	42	43	44	45															
A.	2	3	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	3	2	3	2	1	1	3	4	2	4	3	3	4	4	3	3	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	2	4	2	2	2	2	3	1	1	4	2	3	3	3	4	2	1	3
Q.	86	87	88	89	90															
A.	1	3	4	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	3	3	3	1	2	4	4	1	1	2	4	4	2	3	2	4	3	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	3	3	3	3	2	2	1	4	1	2	3	3	1	4	1	1	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	4	3	3	3	4	2	2	2	3	3	3	3	2	1	3	4	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	4	3	3	3	3	1	2	1	3	3	2	3	3	4	2	3	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	4	2	4	2	4	2	4	4										

SOLUTIONS

PHYSICS

1)

Explanation Question: We have to choose the incorrect option
Concept: This question is based on vectors
Solution:

$$\left|\vec{A}\right| = \sqrt{3^2 + 4^2} = 5$$

$$\left|\vec{B}\right| = \sqrt{6^2 + 8^2} = 10$$

$$\vec{A}.\vec{B} = \left(3\hat{i} + 4\hat{j}\right).\left(6\hat{i} + 8\hat{j}\right)$$

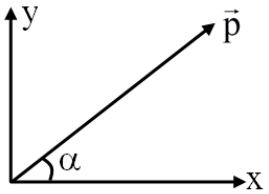
$= 18 + 32 = 50$

$$\frac{\left|\vec{A}\right|}{\left|\vec{B}\right|} = \frac{5}{10} = \frac{1}{2}$$

Final Answer: Option (4)

2)

Explain Question : In this question, we are asked to find the angle that the vector $3\hat{i} + 4\hat{j}$ = makes with the x-axis.
Concept : This question is based on direction of a vector.
Solution : For a vector, $\vec{P} = p \times \hat{i} + p_y\hat{j}$**.** the angle 'α' with the x-axis is given by



$$\tan\alpha = \frac{P_y}{p_x}$$

$$\tan\alpha = \frac{4}{3}$$

$\alpha = 53°$

Final answer : Option (4)

3)

Explain Question :
Find the value of n such that three given vectors are coplanar.

Concept:
For coplanar vectors, the scalar triple product is zero.

Solution:
The condition for coplanarity is that the determinant of the vector components must be zero:

$$\begin{vmatrix} 1 & -2 & 3 \\ 0 & n & -3 \\ 4 & 2 & 2 \end{vmatrix} = 0$$

$$1(2n-(-6))-(-2)(0-(-12))+3(0-4n)=0$$
$$2n+6+24-12n=0$$
$$-10n+30=0 \Rightarrow n=3$$

Final answer : Option (3)

4) **Question Explanation:** In this question, we have to find the unit vector perpendicular to both given vectors say $\vec{A}$ and $\vec{B}$.

Concept: Application of cross product→ Unit vector.

Solution:
A unit vector $\hat{n}$ perpendicular to $\vec{A}$ and $\vec{B}$ is given by

$$\hat{n}=\frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$$

Let the

$$\vec{A}=2\hat{i}+2\hat{j}-\hat{k}$$
$$\vec{B}=6\hat{i}-3\hat{j}+2\hat{k}$$
$$\vec{A}\times\vec{B}=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 6 & -3 & 2 \end{vmatrix}$$
$$\vec{A}\times\vec{B}=\hat{i}(4-3)-\hat{j}(4-(-6))+\hat{k}(-6-12)$$
$$\vec{A}\times\vec{B}=\hat{i}-10\hat{j}-18\hat{k}$$
$$|\vec{A}\times\vec{B}|=\sqrt{1^2+(-10)^2+(-18)^2}=\sqrt{425}=5\sqrt{17}$$
$$\hat{n}=\frac{\hat{i}-10\hat{j}-18\hat{k}}{5\sqrt{17}}$$

Final answer: Option(3)

5) $$\cos\theta=\frac{1}{(1)\left(\sqrt{3}\right)}=\frac{1}{\sqrt{3}}$$

6)

Explain Question:
Identify the vector (s) perpendicular to the given 2D vector $(a\cos\theta)\hat{i}+(b\sin\theta)\hat{j}$.

Concept:
Orthogonality condition $(\vec{A}\cdot\vec{B}=0)$ and 3D orientation.

Solution:

Let $\overrightarrow{A} = (a\cos\theta)\hat{i} + (b\sin\theta)\hat{j}$

For option 1: $\overrightarrow{A} \cdot \overrightarrow{B}_1 = (a\cos\theta)(b\sin\theta) + (b\sin\theta)(-a\cos\theta) = 0$

For option 2: $\overrightarrow{A} \cdot \overrightarrow{B_2} = (a\cos\theta)\frac{\sin\theta}{a} - (b\sin\theta)\frac{\cos\theta}{b} = \sin\theta\cos\theta - \sin\theta\cos\theta = 0$

For option 3: $5\hat{k}$ is perpendicular to any vector in the xy-plane

Since all options satisfy the perpendicularity condition, all are correct

Final answer: Option (4)

7) **Question Explanation:**

In this question we have to find the projection of $\vec{A}$ and $\vec{B}$.

Concept:

Projection of a vector on another vector (application of dot product)

Solution:

Projection of $\vec{A}$ and $\vec{B}$ is given by $= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|}$ given here

$\vec{A} = \hat{i} - \hat{j}$ & $\vec{B} = -\hat{i} + \hat{j}$

$|\vec{B}| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$

$\therefore$ Projection of $\vec{A}$ on $\vec{B} = \frac{(\hat{i} - \hat{j}) \cdot (-\hat{i} + \hat{j})}{\sqrt{2}}$

Projection of $\vec{A}$ on $\vec{B} = \frac{(1 \times -1) + (-1 \times 1)}{\sqrt{2}} = -\sqrt{2}$

Final Answer: Option (2)

8) **Explain the Question :** In this question, we need a vector that has direction same as $\vec{Q} = \hat{i} + \hat{j}$.

Magnitude same as $\vec{P} = 3\hat{i} + 4\hat{j} + 5\hat{k}$.

Concept : This question is based on magnitude of vector and unit vector.

Solution : A vector with magnitude $|\vec{P}|$ and direction of $\vec{Q}$ is

$\vec{R} = |\vec{P}| \times \frac{\vec{Q}}{|\vec{Q}|}$

Step-1

Magnitude of $\vec{P}$

$|\vec{P}| = \sqrt{(3)^2 + (4)^2 + (5)^2} = \sqrt{50} = 5\sqrt{2}$

Step-2

Magnitude of $\vec{Q}$

$|\vec{Q}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$

Step-3 : Unit vector along $\vec{Q}$

$$\hat{Q} = \frac{\vec{Q}}{|\vec{Q}|} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$$

Step-4 : Required vector, $\vec{R} = |\vec{P}| \times \frac{\vec{Q}}{|\vec{Q}|}$

$$\vec{R} = 5\sqrt{2} \times \frac{(\hat{i}+\hat{j})}{\sqrt{2}} = 5(\hat{i}+\hat{j})$$

Final Answer : Option (2)

9) $\vec{A}$ = i + 2j + 3k, $\vec{B}$ = 3i – 2j + k

$$\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & -2 & 1 \end{vmatrix} = 8i + 8j - 8k$$

Area of parallelorgram = $\left|\vec{A} \times \vec{B}\right| = 8\sqrt{3}$

10)

Substituting $\vec{r} = (A\cos\omega t\,\hat{i} + A\sin\omega t\,\hat{j})$

in $\vec{v} = \frac{d\vec{r}}{dt}$ we have

$$\vec{v} = A\frac{d}{dt}(\cos\omega t)\hat{i} + A\frac{d}{dt}(\sin\omega t)\hat{j}$$

$$= -A\omega\sin\omega t\,\hat{i} + A\omega\cos\omega t\,\hat{j}$$

At t = 0, $v = -A\omega\sin 0\hat{i} + A\omega\cos 0\,\hat{j} = A\omega\hat{j}$

11)

$x = at^2 - bt^3$

$v = 2at - 3bt^2$

$a = 2a - 6bt = 0 \Rightarrow t = \frac{a}{3b}$

12) $\boxed{S \propto U^2}\ \frac{8}{S_2} = \frac{30 \times 30}{60 \times 60} \Rightarrow S_2 = 32\,m$

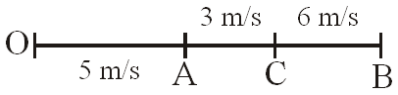
13)

For quarter of circle; Distance = 2πR/4 = πR/2

Displacement = $\sqrt{2}R$

$$\therefore \frac{\text{Distance}}{\text{Displacement}} = \frac{\pi R/2}{\sqrt{2}R} = \frac{22}{2 \times 7 \times \sqrt{2}} = \frac{11}{7\sqrt{2}}$$

14)



A to B

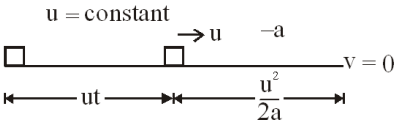
$\langle v \rangle = v' = \frac{2(3)(6)}{3+6} = 4$ m/s

$\Rightarrow$ O → B

$\langle v \rangle = \frac{5+4}{2} = \frac{9}{2} = 4.5$ m/s

15)

Reaction time = 0.4 sec



$\Rightarrow$ $s = (15)(0.4) + \frac{(15)^2}{2 \times 5} = 28.5$ m

16)

Average speed = $\frac{15 \times 2 + 5 \times 18}{2 + 18} = 6$ m/s

17)

at t = 0, v = 0

now $a = 2(t - 1)$ so $\frac{dv}{dt} = 2(t - 1)$

$$\int_0^v dv = \int_0^6 (2t - 2)\,dt$$

$$v = \left[\frac{2t^2}{2} - 2t\right]_0^6 = (6)^2 - 2(6) = 24 \text{ m/s}$$

18)

$0 + \frac{a}{2}(2 \times 8 - -1) = 120 \Rightarrow a = 16$ m/s^2

19) $\xleftrightarrow[v_1]{\frac{2}{5}s}$ $\xleftrightarrow[v_2]{\frac{3}{5}s}$ Avg speed = $\frac{\text{Total dist.}}{\text{Total time}}$

$$= \frac{s}{\frac{2s}{5v_1} + \frac{3s}{5v_2}} = \frac{5v_1v_2}{3v_1 + 2v_2}$$

20)

Distance is zero so object must be at rest.

21)
$$\Rightarrow 10\hat{i} + 40\hat{j} + 40\sqrt{2}\left(\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right)$$
$$\Rightarrow 10\hat{i} + 40\hat{j} - 40\hat{i} - 40\hat{j}$$
$$\Rightarrow -30\hat{i}$$
⇒ 30m in west

22)

$$V_{avg.} = \frac{40 + 80}{2}$$
$V_{avg.}$ = 60 km/hr

23)

$x = 2t^2 - t - 1$
$v = \frac{dx}{dt} = 4t - 1$
x = 0 at t = 1
v at x = 0 or t = 1
= 3 m/s

24)

because initial & final positions are same so the displacement is zero.
$$v_{avg} = \frac{r + \frac{\pi r}{2} + r}{10/60} = \frac{2 + \frac{\pi}{2}}{1/6} = 21.4\,km/hr$$

25)

Question Explanation : In this question we are given with the equation of displacement i.e. $\sqrt{x} = t + 3$ we need to analyze the velocity from this displacement time relation.
Concept : This question is based on Application of calculus for general motion.
Solution :- Given here
$\sqrt{x} = t + 3$ ... (1)
squaring the equation -1 on both side we get
$\Rightarrow x = (t + 3)^2$
$\Rightarrow$ velocity $v = \frac{dx}{dt} \Rightarrow \frac{d}{dt}(t + 3)^2$
$\Rightarrow v = 2(t + 3) = 2t + 6$
$\Rightarrow v = 2t + 6$
comparing with the equation of straight line i.e. y = mx + c
we can see that velocity varies linearly with time 't' ($v \propto t$) and if we put t = 0
we get v = 6m/s which is the initial velocity.
Final Answer : Option (2)

26)

$s = t^3 - 3t^2 + 2$

$\Rightarrow \frac{ds}{dt} = 3t^2 - 6t$

and acceleration $\frac{d^2s}{dt^2} = 6t - 6$

when acceleration is zero, t = 1 sec. Hence, displacement of the particle at t = 1 sec (i.e., when acceleration becomes zero) is,

$s = t^3 - 3t^2 + 2$

$= 1 - 3 + 2 = 0$ m.

27)

Explain Question : Asking about the velocity & displacent equation as the function of time.
Concept : This question is based on application of calculus for general motion.
Explanation :

$$\frac{dv}{dt} = a = kt \Rightarrow \int_0^v dv = \int_0^t ktdt$$

$v = \frac{1}{2} kt^2$, $\frac{ds}{dt} = v = \frac{1}{2}kt^2$

$$\int ds = \frac{k}{2}\int_0^t t^2dt \Rightarrow s = \frac{kt^3}{6}$$

Final answer : 4

28)

Explain Question : To find how long would the particle travel before coming to rest ?
Concept : This question is based on calculus for general motion.
Explanation : $V = \frac{dx}{dt} = 12 - 3t^2 = 0$

$t^2 = \frac{12}{3} = 4$

$t = +2$ sec

$x_f = 20 + 12(2) - (2)^3$

$= 20 + 24 - 8 = 36$ m

$x_i = 20$ (t = 0)

$x_f - x_i = 36 - 20 = 16$m

Final answer : 1

29)

Explain Question: Find the distance a particle travels in the fourth second of its motion, given its initial velocity and constant retardation.
Concept: This question is based on uniformly accelerated linear motion
Formula :

$s_{n^{th}} = u + \frac{a}{2}(2n-1)$

Solution:

$S_{n^{th}} = u - \frac{a}{2}(2n-1)$

$S_{n^{th}} = 10 - \frac{2}{2}(2(4)-1)$

$S_{n^{th}} = 10 - 7$

$S_{n^{th}} = 3m$

Final answer: (1)

30) $v^2 = u^2 + 2as$

$400 - 100 = 2(a)(135)$

$300 = 270a$

$a = \frac{300}{270}$

$= \frac{10}{9} m/s^2$

$v = u + at$

$20 = 10 + \left(\frac{10}{9}\right)t$

$t = 9 \text{ sec.}$

31) Acceleration $= \frac{dv}{dt} = 0.2t$ which is not constant.

32) Average velocity

$= \frac{\text{Displacement}}{\text{Time taken}} = \frac{y(t=3) - y(t=0)}{3-0}$

$= \frac{2(3)^2 + 3(3) + 4 - 4}{3} = 9 \text{ m/s}$

33) Average velocity $= \frac{2 \times 1}{1} = \frac{2}{1} = 2m/s$

34) $a = \frac{Vdv}{dx}$

$a = (3x^2 - 2x)(6x - 2)$

at $x = 2$

$a = 80 \text{ m/sec}^2$

35) From Ist equation of motion $\vec{v} = \vec{u} + \vec{a}t$

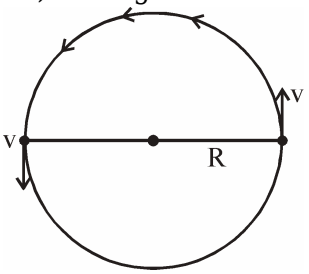
We have

$\vec{v} = (2\hat{i} + 3\hat{j}) + (0.3\hat{i} + 0.2\hat{j})(10) = 5\hat{i} + 5\hat{j}$

Therefore

$|\vec{v}| = \sqrt{5^2 + 5^2} = 5\sqrt{2} \text{ units}$

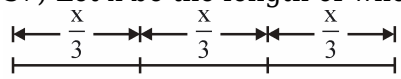
36) Average acc. = $\frac{\Delta v}{t} = \frac{2v}{\left(\frac{\pi R}{v}\right)} = \frac{2v^2}{\pi R}$



Putting v = 8 m/sec
R = 6 m

we get $a_{avg.} = \frac{64}{3\pi}\ m/sec^2$

37) Let x be the length of whole journey.



Average velocity = $\frac{\text{Total displacement}}{\text{Total time taken}}$

$= \frac{x}{\frac{x/3}{2} + \frac{x/3}{3} + \frac{x/3}{6}} = \frac{1}{\frac{1}{6} + \frac{1}{9} + \frac{1}{18}}$

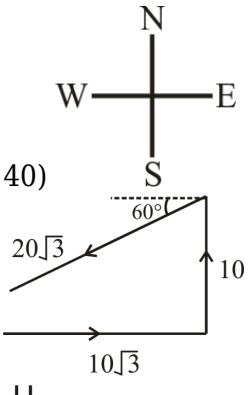
$= \frac{18}{3+2+1} = 3\ m/s$

38) $x = \frac{n^2}{2n--1} \Rightarrow x = \frac{100}{20-1} = \frac{100}{19} \simeq 5.2$
So, bullet is stopped by 6th plank.

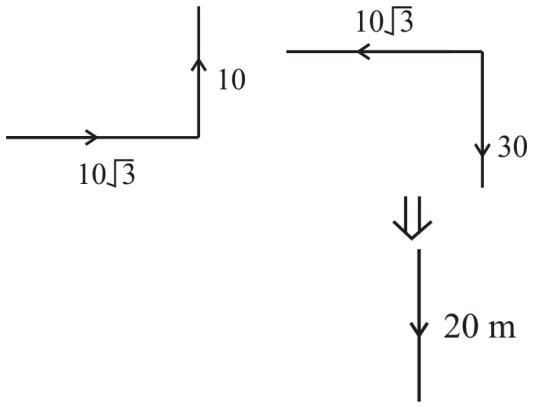
39) In one complete forward & backward cycle the men moves 2m ahead in 6 sec
After moving 6m ahead he will fall down without coming back
total time = Time to cover 6m+Time to cover 4m

$= \frac{6}{2} \times 6 + 4 = 22\ sec$

40)



⇓



41)

Question Explanation :
Find the final distance of an insect from the origin after it moves in two stages with different constant velocities.

Concept:
This question is based on vector addition of displacements.

Formula :
Displacement vector: $s = v_t$
Total displacement = $s_1 + s_2$
Magnitude of vector $\vec{s} = x\hat{i} + y\hat{j} : |\vec{s}| = \sqrt{x^2 + y^2}$

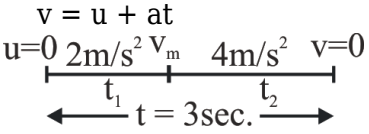
Solution :
$\Delta\vec{r} = \vec{v}_1\Delta t_1 + \vec{v}_2\Delta t_2$
$= \left(2\hat{i} + 3\hat{j}\right)(1) + \left(2\hat{i} + 2\hat{j}\right)\left(\frac{1}{2}\right)$
$= \left(3\hat{i} + 4\hat{j}\right)$ m
Magnitude of the total displacement (distance from origin): $|\Delta\vec{r}| = \sqrt{(3)^2 + (4)^2} = \sqrt{25} = 5$

Final answer:
Option (2)

42)

Displacement = 2m
Distance = 2π + π = 3π m
Ratio = $\frac{3\pi}{2}$

43) From,
$v = u + at$



for t_1

$v_m = 2t_1$

or $t_1 = \frac{v_m}{2}$ .....(i)

Similarly, $t_2 = \frac{v_m}{4}$

As given,

$t_1 + t_2 = t$

or $t_1 + t_2 = 3$

or $\frac{v_m}{2} + \frac{v_m}{4} = 3$ [v_m max speed]

$\Rightarrow$ $v_m = 4$ m/s

44)

$v = (180 - 16x)^{1/2}$

$a = \frac{(v)dv}{dx}$ $a = \frac{(v)dv}{dx}$

$a = (180-16x)^{1/2} \cdot \frac{d}{dx}(180-16x)^{1/2}$

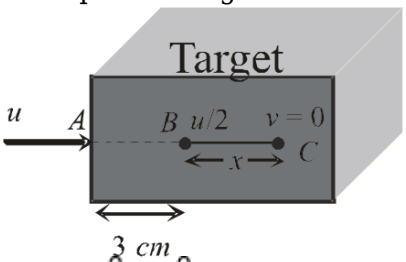
$a = \frac{1}{2}\left[\frac{(180-16x)^{1/2}}{(180-16x)^{1/2}}\right]$ $[0-16]$

$= -8$ m/s^2

45)

Let initial velocity of the bullet = u

After penetrating 3 cm its velocity becomes $\frac{u}{2}$



From $v^2 = u^2 - 2as$

$\left(\frac{u}{2}\right)^2 = u^2 - 2a(3) \Rightarrow 6a = \frac{3u^2}{4} \Rightarrow a = \frac{u^2}{8}$

Let further it will penetrate through distance x and stops at point C.

For distance BC, $v = 0, u = u/2, s = x, a = u^2/8$

From

$v^2 = u^2 - 2as \Rightarrow 0 = \left(\frac{u}{2}\right)^2 - 2\left(\frac{u^2}{8}\right).x$

$\Rightarrow x = 1$ cm.

CHEMISTRY

46) **Question Explanation:**
The question asks to convert length from millimeter into meter.

Concept:
Unit Conversion.

Solution:
1 milli = 10^{-3}
1 milli meter = 10^{-3} meter
10^{-2} mm = $10^{-2} \times 10^{-3}$ m
10^{-2} mm = 10^{-5} m

Final Answer: Option (2)

47)

Question Explanation : We need to find incorrect unit conversion relation.

Concept : Inter conversion of units.

Solution :

Relation between erg and Joule –

1J = 10^{7} erg

In option (3) :-

1 erg = 10^{7} J (Incorrect)

Final Answer: option (3)

48)

We know 1 cm^3 = 1 mL so,
4 cm^3 = 4 mL

49) $\frac{w}{M_w} = \frac{N}{N_A}$
$\frac{16}{16} = \frac{N}{N_A}$
Molecule = N_A
$= 6.02 \times 10^{23}$

50) $\frac{w}{M_w} = \frac{V}{22.4}$
Here M_w = 28

51)

Explaining the Question :- What is the mass of 3.01×10^{22} molecules of ammonia (NH_3)

Concept:- This question is based on Mass = (given molecules / Avogadro number) × Molar mass

Solution :- Number of moles = $(3.01 \times 10^{22}) / (6.022 \times 10^{22}) = 0.05$ mol
Mass = $0.05 \times 17 = 0.85$ g
Final Answer :- 2

52)

M = 2VD
$= 2 \times 44 = 88$
$(CO_2)_x = 88$
$44 \times x = 88$
$x = 2$

53) **Explain Question:** We must find which gas samples have the same number of molecules.
Concept: This question is based on Mole= Given mass/Molar mass
Solution:
Equal moles ⇒ equal number of molecules
1. 16 g O_2: moles = 16/32 = 0.5
14 g N_2: moles = 14/28 = 0.5 (equal)
2. 8 g O_2: moles = 8/32 = 0.25 mol
22 g CO_2: moles =22/44 = 0.5 mol (not equal)
3. 28 g N_2: moles = 28/28 = 1 mol
22 g CO_2: moles = 22/44 = 0.5 mol (not equal)
4. 32 g O_2: moles = 32/32 = 1 mol
32 g N_2: moles = 32/28 ≈ 1.14 mol (not equal)
16 g O_2 and 14 g N_2 have equal moles
Final Answer: Option 1

54)

Total no of electrons present = no of moles $\times$ $N_A \times$ electrons carried by single species
$= \frac{48}{24} \times N_A \times 10$
$= 20 \times N_A$

55)

Explaining the Question :- Which of the following has minimum number of atoms of oxygen?

Concept:- This question is based on Moles = given mass / molar mass, Number of particles = moles × Avogadro's number (6.022×10^{23}).

Solution :-
(1) 10 mL H_2O (□)
Density ≈ 1g/mL → mass = 10 g
Oxygen atoms = $0.556 \times 6.02 \times 10^{23} = 3.35 \times 10^{23}$ atoms.
(2) 0.1 mL V_2O_5 (s)
Each molecule has 5 O atoms.
Oxygen atoms = $0.1 \times 5 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$atoms.
(3) 12 g O_3 (g)
Molar mass = 48/g mol → moles = 12/48 = 0.25 mol.
Each molecule has 3 O atoms
Oxygen atoms = $0.25 \times 3 \times 6.022 \times 10^{23} = 4.52 \times 10^{23}$ atoms.
(4) 12.04×10^{22} molecules of CO_2
Each molecule has 2 O atoms.
Oxygen atoms = $12.4 \times 10^{22} \times 2 = 2.41 \times 10^{23}$ atoms.
Final Answer :- 4

56)

Molecular mass = 18 amu = $\frac{18}{N_A}$ g = $\frac{18}{6.022 \times 10^{23}}$

57)

Initial mole = $\frac{120}{100} = 1.2\,\text{mol}$
removed moles = $\frac{6 \times 10^{22}}{6 \times 10^{23}} = 0.1\,\text{mol}$
remaining moles = 1.2 – 0.1
= 1.1 mole

58)

Molecule ratio = mole ratio = $\frac{1}{32} : \frac{4}{28} = \frac{1}{32} : \frac{1}{7}$

59) $\text{Actual mass of } CO_2 = \frac{44}{6.023 \times 10^{23}}\text{g}$

60) **Explain Question:** We are asked which sample contains the maximum number of atoms for 16 g of oxygen in different forms: O, O_2, and O_3.
Concept: This question is based on Number of moles = mass / molar mass, Number of atoms = moles × Avogadro's number × atoms per molecule
Solution:
1. 16 g O (atomic oxygen): Molar mass = 16 g/mol
Moles = 16 / 16 = 1 mol
Atoms = $1 \times 6.022 \times 10^{23} = 6.022 \times 10^{23}$ atoms
2. 16 g O_2 (molecular oxygen): Molar mass = 32 g/mol

Moles = 16 / 32 = 0.5 mol
Atoms = $0.5 \times 6.022 \times 10^{23} \times 2 = 6.022 \times 10^{23}$ atoms
3. 16 g O_3 (ozone): Molar mass = 48 g/mol
Moles = 16 / 48 ≈ 0.333 mol
Atoms = $0.333 \times 6.022 \times 10^{23} \times 3 \approx 6.022 \times 10^{23}$ atoms
All will have same number of atoms
Final Answer: Option 4

61) $\frac{w}{M_w} = \frac{N}{N_A}$

$\frac{360 \times 10^{-3}}{180} = \frac{N}{N_A}$

Molecule = $2 \times 10^{-3} \times 6 \times 10^{23}$
'O' atom = $2 \times 10^{-3} \times 6 \times 10^{23} \times 4$
$= 4.8 \times 10^{21}$

62) $4 = \frac{32 \times 1 \times 100}{\text{min. mol. wt}}$

min. mol. wt = $\frac{32 \times 100}{4} = 800$

63) $M = \frac{W \times 1000}{M_w \times V_{ml}}$

$0.1 = \frac{0.4 \times 1000}{40 \times V_{ml}}$

V_{ml} = 100 ml

64)

$2Mg + O_2 \rightarrow 2\ MgO$
1.5 mole

$n_{Mg} = \frac{2}{1} \times 1.5 = 3$

65)

Question Explanation : We need to find required volume of O_2 for complete combustion of 10L of CH_4 gas.
Concept : This question is based on Stoichiometry based concept
Solution :
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
1L CH_4 reacts with 2 L of O_2 gas
10L CH_4 reacts with 20 L of O_2 gas
Required volume of O_2 gas = 20 L
Final Answer : Option (3)

66) 2 mole of Ca produces 2 mole of CaO
$= 2 \times (40 + 16) = 2 \times 56 = 112\ g$

67)

Moles of solute = $\frac{6.02 \times 10^{20}}{6.02 \times 10^{23}} = 10^{-3}$

$M = \frac{\text{moles of solute}}{V_{(ml)}\text{solution}} \times 1000$

$= \frac{10^{-3}}{100} \times 1000$

$M = 10^{-2}$

68) **Question Explanation:** What volume of CO_2 will be formed by complete combustion of 2 □ of C_3H_8?
Concept: Stoichiometry, volume-volume relationship
Solution:
Combustion reaction: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
Volume ratio of gases = Mole ratio (Avogadro's law).
1 volume of C_3H_8 gives 3 volumes of CO_2.
From the balanced equation: 1 □ $C_3H_8 \rightarrow 3$ □ CO_2
Therefore, 2 □ $C_3H_8 \rightarrow 2 \times 3 = 6$ □ CO_2
Final Answer: Option: 2

69) $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2\ H_2O$
1 L 3 L
□ 10 L 30 L
and volume of air = 30 × 5 = 150 L
(□ Air contains approximately 20% of O_2 by volume)

70)

Explaining : -
Concept : This question is based on
Solution :

Moles Al = $\frac{54}{27} = 2$ mol

Moles Cl_2 = $\frac{71}{71} = 1$ mol

Required ratio = Al:Cl_2 = 2 : 3 → but actual ratio = 2 : 1
Cl_2 is limiting reagent.
Final Answer : (2)

71)

Explaining : -
Concept : This question is based on

Solution :
Assume 100 g compound :
S = 40 g → $\frac{40}{32}$ = 1.25 mol ;
O = 60 g → $\frac{60}{16}$ = 3.75 mol
Ratio = 1.25 : 3.75 = 1 : 3
Empirical formula = SO_3

72)

Explaining : -
Concept : This question is based on
Solution :
C = $\frac{85.7}{12}$ = 7.14
H = $\frac{14.3}{1}$ = 14.3
Ratio = 1 : 2
Empirical formula = CH_2
Empirical mass = 14, molecular mass = 2 × 28 = 56
n = $\frac{56}{14}$ = 4
Molecular formula = $(CH_2)_4 = C_4H_8$
Final Answer : (2)

73)

$2A + 3B \rightarrow 2C$
4mol 12 mol
A is LR
2 mole of A gives 2 mol of C
4 mol of A gives 4 mol of C

74)

$CaCO_3 \rightarrow CaO + CO_2$
$\frac{40 \times 10}{100}$
w = 4g ⇒ n = $\frac{4}{100}$
$n_{CO_2} = \frac{1}{1} \times \frac{4}{100}$
$\frac{w}{44} = \frac{4}{100} \Rightarrow w = 1.76\ g$

75)

SO_2 is LR
Answer is "0 mol, 10 mol"

76) **Explanation question:**
We need to find mole fraction of A if 2 moles of A is present in 3 moles of B.
Concept:
mole fraction.
Solution:
A → moles = n_A
B → moles ⇒ n_B
Mole fraction of A ⇒ $X_A = \frac{n_A}{n_A + n_B}$
$X_A = \frac{2}{2+3} = \frac{2}{5} = 0.4$
$X_A = 0.4$
Final Answer: option (1)

77)

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2O$

$V = 11.2\ L$

$n = \frac{11.2}{22.4}$

$n = 0.5$

$n_{Fe} = \frac{3}{4} \times 0.5 = \frac{1.5}{4}$

$\frac{w}{56} = \frac{1.5}{4} \Rightarrow w = \frac{1.5 \times 50}{4}$

$= 21\ g$

78) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

Mole of $O_2 = \frac{224}{22.4} = 10$

2mole of $Pb(NO_3)_2$ Produce 1 mole of O_2

x mole of $pb(NO_3)_2$ produce 10 mole of O_2

$x = 2 \times 10 = 20$

79) no.of moles of water $= \frac{54}{18} = 3$
no.of neutrons $= 3 \times 8 \times N_A = 24\ N_A$.

80) $2H_2 + O_2 \rightarrow 2H_2O$
$\frac{1}{2}$mole $\frac{1}{8}$mole
$\therefore \frac{1}{4}$mole H_2O is formed

81) Weight of oxygen = 40g
Weight of metal in metal oxide = 100 – 40 = 60g

No. of moles of metal $= \frac{60}{24} = 2.5$

No. of moles of oxygen $= \frac{40}{16} = 2.5$

simplest ratio of metal and oxygen = 1 : 1

Therefore MO

82)

$2A\square + \frac{3}{2} O_2 \rightarrow Al_2O_3$

mole of $A\square = \frac{9}{27} = \frac{1}{3}$

mole of $O_2 = \frac{3}{2 \times 2} \times \frac{1}{3} = \frac{1}{4}$

$\frac{1}{4} = \frac{w}{32} \Rightarrow w = 8g$

83) mass fraction of glucose $= \frac{18}{18+162} = \frac{18}{180} = 0.1$

84)

Conceptual

85)

Theory based.

86)

Question Explanation: We must identify the sign of enthalpy change for an endothermic reaction.

Concept: Endothermic reactions

Solution:

Endothermic reactions absorb heat from surroundings, so enthalpy of products is greater than reactants. Hence ΔH is positive.

Heat absorbed ⇒ H(products) > H(reactants)

Therefore ΔH > 0.

Final Answer: Option 1

87)

Explain Question : The question asks for the defining characteristic of a chemical reaction at equilibrium from the given option.

Concept : This question is based on characteristics of chemical equilibrium.

Solution :
At equilibrium
Rate of forward reaction = Rate of backward reaction
$r_f = r_b$
Final answer : 3

88)

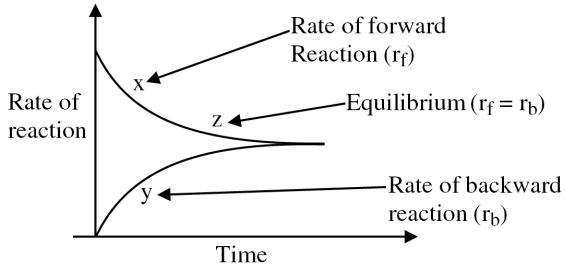
Theory based

89)

Equilibrium is dynamic in nature.

90)

Explain the Question : With the help of given graph we need to identify x, y and z.
Concept : This question is based on equilibrium state.
Solution :



x ⇒ for forward reaction
y ⇒ for backward reaction
z ⇒ for equilibrium
Final Answer : 2

BIOLOGY

91)

NCERT Pg. # 46

92)

NCERT Pg. # 50, 51

93)

NCERT XI Pg.# 46

94) NCERT- XI, Page No.- 48

95) NCERT, Pg. # 46

96)

NCERT Pg. # 47

97)

NCERT Pg. # 49

98)

NCERT Pg. # 49

99)

NCERT Pg. # 46, 47

100)

NCERT XI, Page # 47

101)

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102)

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103)

NCERT XI, Page # 49

104)

NCERT Pg. # 49

105)

NCERT Pg. # 47, 48

106)

NCERT Pg. # 48

107)

NCERT Pg. # 47, 48

108)

NCERT Pg. # 49

109)

NCERT Pg. # 46

110)

NCERT Pg. # 47

111)

NCERT Pg. # 47, 48

112)

NCERT Pg. # 51

113)

NCERT Pg. # 48

114)

NCERT Pg. # 49

115)

NCERT Pg. # 49, 50

116)

NCERT Pg. # 48

117)

NCERT Pg. # 47

118)

Assertion: Acraniata is a group of organisms which do not have distinct cranium. **TRUE.**

Reason: It includes small marine forms without head. TRUE. Acraniates don't have a clearly defined head region.

Explanation:
While both the assertion and the reason are true, the reason isn't the direct explanation for the assertion. The primary reason Acraniata don't have a distinct cranium is that they lack the evolutionary development of a bony or cartilaginous skull that encloses and protects the brain. The size and lack of a distinct head are more general characteristics of their body plan, not the direct cause for the absence of a cranium.

Final Answer : option (2) **If both the assertion and reason are true but the reason is not a correct explanation of the assertion.**

119)

Question Explanation :
Identify the group of animals based on given structural features.

Concept : Chondrichthyes

Solution :
Cartilaginous endoskeleton, ventral mouth, placoid scales, persistent notochord, and absence of air bladder are characteristic features of Chondrichthyes.

Final Answer : Option (1)

120) NCERT Pg. No. 47, 49, 50

121) **Explaining :**
Concept :
Solution : Larynx is referred to as voice box/sound box for sound production.
Final Answer : (1)

122) **Explaining :**
Concept :
Solution : The tracheae, primary, secondary and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.
Final Answer : (2)

123) **Explaining :**
Concept :
Solution : We have two lungs which are covered by a double layered pleura, with pleural fluid between them. It reduces friction on the lung-surface.
Final Answer : (3)

124) **Explaining :**

Concept :
Solution : This includes RV and ERV (2500ml).
Final Answer : (3)

125) **Explaining :**
Concept :
Solution : Most oxygen is carried by haemoglobin and most carbon dioxide is carried as bicarbonate ions.
Final Answer : (1)

126) **Explaining :**
Concept :
Solution : The correct sequence of stages of respiration is:
Breathing → Gaseous transport → Tissue respiration → Cellular respiration.
Final Answer : (2)

127)

NCERT Pg. # 183

128)

NCERT Pg. # 183

129)

Question Explanation: The question asks how partial pressures of O_2 and CO_2 in atmospheric air compare to alveolar air.
Concept: Respiratory physiology – Partial pressures of gases
Solution: Atmospheric air has $PO_2 \approx 159$ mm Hg and $PCO_2 \approx 0.3$ mm Hg. Alveolar air has $PO_2 \approx 104$ mm Hg and $PCO_2 \approx 40$ mm Hg. Therefore, PO_2 in atmospheric air is higher than in alveolar air, and PCO_2 in atmospheric air is lower than in alveolar air.
Final Answer: Option 2

130)

NCERT Pg. # 184

131)

NCERT Pg. # 185, 186

132)

Explain Question : This question asking about regulation of breathing and the relative roles of pO_2 vs pCO_2/H^+.

Concept : This question is based on regulation of respiration.
Explanation : Peripheral/central chemoreceptors primarily respond to increased pCO_2 and H^+; hence O_2 has no role in routine regulation, and the reason correctly explains the assertion.
Final answer : 1

133)

NCERT Pg. # 189

134)

NCERT Pg. # 186, 187

135) **Question Explanation:**
This question is asking about the effect of increased blood CO_2 levels on the oxygen–hemoglobin (O_2–Hb) dissociation curve at 37°C.

Concept :
Breathing and Exchange of Gases (Bohr effect)

Explanation:
An increase in CO_2 lowers blood pH, decreasing Hb's affinity for O_2. This causes the O_2–Hb dissociation curve to shift to the right, facilitating oxygen release to tissues.

Final Answer: Option 3

136)

NCERT-XI, Pg. # 123

137) **Solution:** Telocentric chromosomes have a **terminal centromere**, not an absent one.
Final Answer : (4)

138) **Solution:** 80S ribosomes = 60S (large) + 40S (small) subunits.
Final Answer : (2)

139) **Explanation:** According to electron microscopic studies, the internal core that contains the microtubules is termed the axoneme.
Final Answer : (2)

140) **Explanation: Kinetochores.** The text explicitly mentions that kinetochores are disc-shaped structures present on the sides of the centromere.
Final Answer : (2)

141) NCERT-XI, Pg. # 100

142) NCERT-XI, Pg. # 100

143) **Solution :** Axoneme of cilia/flagella shows 9 doublets + 2 central microtubules.
Final Answer : (3)

144) **Solution :** Ribosomes are known as the 'protein factories' because they are the sites of protein synthesis, assembling amino acids into polypeptides.
Final Answer : (3)

145) NCERT-XI, Pg. # 101

146) NCERT-XI, Pg. # 101

147) NCERT Pg. no. 99 fig. 8.10 (b)

148) NCERT-XI, Pg. # 101

149) **The correct answer is: 4. Both (1) & (2)**

Explanation:
Double-stranded circular DNA is found in both **mitochondria** and **chloroplasts** of plant cells. These organelles have their own DNA, similar to bacteria, supporting the endosymbiotic theory.

150)

NCERT, Pg. # 98

151) Module Pg. # 193

152)

NCERT Pg. # 100, 101

153) **Question Explanation :**
Which of the following are functions of cytoskeleton?

Concept :
This question is based on Cell – Cytoskeleton (Structure & Functions)

Solution :
The cytoskeleton is a network of microtubules, microfilaments, and intermediate filaments present in the cytoplasm of eukaryotic cells.
Functions:

• Motility → helps in movement (cilia, flagella, intracellular transport).
• Mechanical support → provides strength to cell contents.
• Maintenance of shape → gives shape and polarity to the cell.

Final Answer :
Option (4)

154)

NCERT-XI, Pg. # 96, 97

155)

NCERT-XI, Pg. # 101

156)

Similar Chromosomes

157)

Explain Question : Which of the following is true for nucleolus.
Concept : This question is based on Nucleus.
Explanation : The nucleolus is a dense, non-membrane bound structure in the nucleus responsible for ribosomal RNA (rRNA) synthesis and ribosome assembly.
Final answer : 3

158)

NCERT-XI, Pg. # 97, 98

159) **Solution:** Amyloplasts → Starch
Elaioplasts → Oils & fats
Aleuroplasts → Proteins
Final Answer : (2)

160) **Explaining:** Energy metabolism role of mitochondria.
Concept: Mitochondria – ATP synthesis
Solution: ATP synthesis is the basis of the term.
Final Answer: (1)

161) **Explaining:** Structural modification of mitochondria.
Concept: Mitochondrial membranes
Solution: Cristae arise from inner membrane.
Final Answer: (3)

162) **Explaining:** Components and role of cytoskeleton.
Concept: Cytoskeleton
Solution: It includes microfilaments and intermediate filaments also.
Final Answer: (3)

163) Synthesis and storage of lipid

164) NCERT Page No. 133

165)

NCERT-XI, Pg. # 102

166) NCERT (XI) Pg. # 164

167) Module-1 Pg. # 206

168) Question Explanation :
The question asks to identify the most "dramatic" period of the cell cycle.

Concept :
This question is based on M-phase (Mitosis).

Explanation:
"The M phase is the most dramatic period of the cell cycle involving a major reorganisation of virtually all components of the cell."

Final answer : 3

169) **Question Explanation:** Identifying a phase of the cell cycle.
Concept: Cell cycle phases
Solution: The cell cycle consists of two main phases: interphase and the M phase (mitosis). **Interphase** is the period of cell growth and DNA synthesis and is divided into three stages: G_1, S, and G_2. In the **G_1 phase**, the cell grows, synthesizes proteins and RNA, and is metabolically active, but its DNA content does not double. DNA replication takes place in the S-phase.
Final Answer: G_1

170) NCERT Pg. # 164

171)

NCERT-XI, Pg. # 121-123

172) NCERT Pg. # 103

173)

NCERT-XI, Pg. # 121-122, 125-127

174)

SOLUTION —
Interphase is the phase of the cell cycle where the cell prepares for division. It is subdivided into three stages:

A. G_1 (Gap 1) – Cell growth and normal functions.

B. S (Synthesis) – DNA replication occurs.

C. G_2 (Gap 2) – Further growth and preparation for mitosis.

The G_o (G-zero) phase is not technically part of interphase, but it's a resting phase that some cells enter from G1, where they are not actively dividing.

The correct answer is: OPTION(2) G_1, S, G_2.

175) **Question Explanation :**
The question asks for the primary protein component of spindle fibers, which are essential structures for cell division.

Concept:
This question is based on cytoskeleton.

Explanation :
"Spindle fibres are made up of microtubules composed of tubulin."

Final answer: 4

176)

Question Explanation: Chromosome number during mitotic metaphase.

Concept: Mitosis and cell division.

Solution: The number of chromosomes remains the same during metaphase as it was in the parent cell. Chromosomes align at the metaphase plate, but they haven't been pulled apart yet. Each chromosome consists of two sister chromatids, but it is still counted as a single chromosome.

Final Answer: 42

177) NCERT XI, Pg # 123
Metaphase

178) NCERT (XI) Pg. # 169, Fig. 10.3

179) **Question Explanation**: Asking about the event not observed during meiosis I
Concept : Meiosis-I
Solution : During meiosis I Division of centromere is not observed
Final answer : Division in centromere

180) NCERT XI, Page # 168, 169